

Right Coset of a subgroup H in G with respect to $a \in G$

$$Ha = \{ ga \mid g \in H \}$$

Similarly, we can define a left coset:

$$aH = \{ ah \mid h \in H \}$$

All that we discussed with right cosets hold true even for left cosets.

Defn: A subgroup N of G is said to be a normal subgroup of G if $\forall g \in G, n \in N, gng^{-1} \in N$.

Obs: $gn \in Ng$ or $ng^{-1} \in g^{-1}N$

Lemma: N is a normal subgroup of G iff $gNg^{-1} = N$ for every $g \in G$.

Pf: If $gNg^{-1} = N$ for every g then by defn, there is nothing left to prove.

If N is normal, then for every $g \in G, gng^{-1} \in N$ for all $n \in N \Rightarrow gNg^{-1} \subseteq N$.

Since $g \in G, g^{-1}Ng \subseteq N$ as well. $\left. \begin{array}{l} \\ \end{array} \right\} N = g(g^{-1}Ng)g^{-1} \subseteq N$

For all those elements in $g^{-1}Ng$, we get that $g^{-1}(gNg^{-1})g \subseteq N$

Suppose $\exists n, g$ s.t. $gNg^{-1} \not\subseteq N$. Then there exists a elem $\tilde{n}$ s.t. $\tilde{n}$ cannot be expressed as gng^{-1} for any g nor n .

But N is normal. then $\exists \hat{n}$ s.t. $\hat{g}\tilde{n}\hat{g}^{-1} = \hat{n}$
 $\Rightarrow \tilde{n} = \hat{g}^{-1}\hat{n}\hat{g}$

This is a contradiction.

Remark: This does not mean that $gng^{-1} = n$ always.

Lemma: The subgroup N of G is a normal subgroup of G if and only if every left coset of N in G is a right coset of N in G .

Pf: If N is a normal subgroup of G then for every $g \in G$, $gNg^{-1} = N \Rightarrow gN = Ng$.

This is true for every g and thus the correspondence is established.

If every left coset is a right coset of N in G then

$\forall g \in G, \exists g' \in G$ s.t. $gN = Ng'$.

Note that $g \in gN$ and thus $g \in Ng'$. Thus, $\exists n \in N$ s.t. $g = ng'$.

OTOH, g also is in $Ng \Rightarrow Ng' = Ng$.

$\Rightarrow gN = Ng \Rightarrow gNg^{-1} = N$

If H, K are two subgroups of G , let

$$HK = \{x \in G \mid x = hk, h \in H, k \in K\}.$$

Lemma:

HK is a subgroup of G iff $HK = KH$.

Proof: If $HK = KH$: Identity and Associative are trivial.

1. Closure: $\underbrace{hk}_x \cdot \underbrace{h'k'}_y \rightarrow \underbrace{h(kh')}_\exists h''k'' \in HK \text{ s.t. } kh' = h''k'' \cdot k'$

$$\downarrow$$

$$h(h''k'')k' = (hh'')(k''k') \in HK.$$

2. Inverse: $x^{-1} = (hk)^{-1} = k^{-1}h^{-1} \in KH = HK$
 $\Rightarrow x^{-1} \in HK.$

If HK is a subgroup then, we can show that $HK \subseteq KH$
 $KH \subseteq HK.$

Theorem: $|HK| = \frac{|H| \cdot |K|}{|H \cap K|}$ Proof: Count the repetitions.

$$e = h^{-1}(hk)h^{-1} = (h^{-1}h)(k^{-1}k) = (h_1 h_1^{-1})(h_1^{-1} k)$$

$$\text{for all } h_1 \in H \cap K \Rightarrow \text{reps} \geq |H \cap K|$$

If $hk = h'k' \Rightarrow k = h^{-1}h'k' \Rightarrow h^{-1}h' = k^{-1}k' = e$

Note that $HH = H$ for any subgroup H .

$$\Rightarrow NaNb = N(aN)b = N(Na)b = NNab = Nab.$$

Lemma: A subgroup N of G is a normal subgroup if, product of two right cosets of N in G is again a right coset in G .

Pf: Rev. dir: If $NaNb = Nc$ then we need to show that N is a normal subgroup.

Consider $Ng \cdot Ng^{-1} = Nc$ for some c .

Note that $g \in Ng$ and $g^{-1} \in Ng^{-1} \Rightarrow e \in Nc$.

But $e \in N$ and thus $Nc = N$.

$ng \cdot n'g^{-1} = n''$ This is true for all g and n' .

$g n' g^{-1} = n'' \in N$ $\Rightarrow N$ is normal.

Obs: All right cosets $\uparrow$ of a normal subgroup form a group as well.

Defn: G/N : Collection of all right cosets of N in G .

Homomorphism: A mapping ϕ is a homomorphism ~~if~~.

$$\phi: G \rightarrow G'$$

if $\forall a, b \in G$ then $\phi(a)\phi(b) = \phi(ab)$

Ex: $\phi(x) = x$.

$\cdot G = \mathbb{R}$ under $+$, $G' = \mathbb{R}$ under $\times$ $\phi(x) = 2^x$.

$\cdot G = \{0, \dots, n-1\}$ $G' = \{e^{\frac{2\pi i k}{n}}\}$ $k \in \mathbb{Z}$ $\times$.

Lemma: If ϕ is a homomorphism of G into G' then.

1. $\phi(e) = e'$

2. $\phi(x^{-1}) = \phi(x)^{-1}$.

Pf: 1. $\phi(x) \cdot \bar{e}' = \phi(x) = \phi(xe) = \phi(x) \cdot \phi(e)$
 $\Rightarrow \bar{e}' = \phi(e)$.

2. $e' = \phi(e) = \phi(x \cdot x^{-1}) = \phi(x) \cdot \phi(x^{-1}) \Rightarrow (\phi(x))^{-1} = \phi(x^{-1})$.

Lemma: G is a group, N a normal subgroup of G .
and def $\phi: G \rightarrow G/N$ by $\phi(x) = Nx \quad \forall x \in G$. Then
 ϕ is a homomorphism from G to G/N .

Lemma: Let Kernel of ϕ (denoted K_ϕ or $\text{Ker}(\phi)$) is
 $\{x \in G \mid \phi(x) = e'\}$.

Then K_ϕ is a normal subgroup of G .

Pf: If $x, y \in K_\phi$ then so does xy .

$$\phi(xy) = \phi(x) \cdot \phi(y) = e' \cdot e' = e'.$$

If $x \in K$ then so does x^{-1} .

To show that K_ϕ is normal, $\forall g \in G$ we need $gkg^{-1} \in K$.
To see that, look at $\phi(gkg^{-1}) = \phi(g) \cdot e' \cdot \phi(g^{-1}) = e'$.
 $\Rightarrow gkg^{-1} \in K \quad \forall k \in K, g \in G$.